

Mohammed Alyafeai

Applied AI Researcher & Engineer

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Objective

Research-focused AI practitioner specializing in large language models (LLMs), with hands-on experience developing state-of-the-art models and systems. Took ownership of and led individual LLM research initiatives, contributing to model development, fine-tuning, and evaluation, while also building robust tooling and pipelines that translate research results into reliable, production-oriented systems, with a strong focus on Arabic LLMs.

Experience

Senior Associate AI Engineer [TII – Abu Dhabi, UAE]

Dec 2023 – Present

- Took ownership of and led a **tool-calling project** that is not yet released, achieving **promising internal results** through a robust and scalable pipeline.
- Co-developed **Falcon-Arabic**, a **state-of-the-art** Arabic LLM (top in its size category at the time), as part of a two-person team.
- Sole developer of a private **state-of-the-art translation model** specialized in Arabic.
- **Fine-tuned** LLMs for multiple tasks, including translation, tool calling, and structured output generation.
- **Designed and integrated supporting GenAI tools** such as RAG pipelines, web search, and crawlers.
- Built custom benchmarks and **automated evaluation workflows** for LLM **research and experimentation**.
- **Improved model** robustness and performance through data synthesis and augmentation techniques.
- Implemented serverless workflows (Lambda) and **automated** cloud-based processes to support LLM pipelines.
- Conducted research experiments and contributed to **scientific research papers**.
- Implemented **automated** scripts to clean unused cloud infrastructure storage, leading to significant daily cost **savings (>\$5K/day)**.

AI/ML & Backend Engineer [Aixplain – California, US] [remotely]

Feb 2023 – Feb 2024

- Integrated multiple ML models using KServe, exposing them through user-friendly APIs for inference and onboarding.
- Containerized model pipelines to support synchronous and asynchronous execution.
- Configured and deployed services on Kubernetes, supporting scalable model hosting and inference workflows.
- Utilized AWS ECR for container images and AWS S3 for model assets and artifacts.
- Built backend systems leveraging vector databases (Qdrant) for similarity search and retrieval use cases.
- Implemented end-to-end CI/CD pipelines using GitHub Actions and custom scripts to automate solution onboarding.
- Took ownership of a GitHub repository, contributing the majority of features and performing code reviews.
- Conducted extensive SDK testing, fixed bugs, and automated documentation generation using Sphinx.

Data Analyst & Software Engineering [Cyber Gate Defense – Abu Dhabi, UAE]

Sep 2021 – Mar 2023

- Built and automated data ingestion pipelines from multiple SIEM solutions.
- Developed ML, deep learning, and NLP models for security and analytics use cases.
- Automated generation of Power BI reports using Python.
- Developed and maintained a data visualization portal for monitoring customer environments.
- Managed Elasticsearch clusters, indexed data, and visualized insights using Kibana.
- Parsed logs, JSON, and YAML data using Logstash, and implemented watchers for device monitoring.
- Documented technical specifications and maintained clean, reproducible codebases.

Education

Al Ain University – Abu Dhabi Campus

Bachelor on Software Engineering 2018-2021

GPA: 3.91 (honored)

Projects & Contributions

- **Falcon-Arabic:** State-of-the-art Arabic LLM in its size category, co-developed as part of a two-person team.
- **Tool-Calling System:** Led the development of a tool-calling project with a robust pipeline, delivering strong and promising results compared to existing approaches (not yet released).
- **Private Translation Model:** Developed a private state-of-the-art translation model specialized in Arabic translation.
- **Document Search Engine by Using NLP:** Search Engine that search for an answer from set of documents. [Python, NLP, fine-tuning, Django]
- **IDRF:** A program that combines the AI and GIS to find the best path in the desert. [Python, AI, Graphs, Shortest Path, Sorting]
- **Model for diacritization:** Manage to build and tune a model that diacritize Arabic text depending in UAE dialect. Worked in fine tuning it to get a good accuracy that qualified me to win the first place in competition of Sibaq Lahja. [Python, NLP by using Deep learning]

- **AutoAuditor:** A web app that use LLMs to do security auditing. [LLMs, Python, Django, redis, celery]
- **Chatbot Generator:** A Website which enable user to create his own chatbot and provide the user with API.[Python, vector database, NLP]

Skills

Key Competencies:

- Leading
- Taking initiative
- Creative Thinking
- Fast Learner
- Teamwork

Languages:

- Arabic (Native)
- English (fluent)

AI & Data Analysis Skills:

- Data analysis using Python libraries such as NumPy and pandas.
- Hands-on experience building machine learning and deep learning models using scikit-learn, and PyTorch.
- Experience developing large language models (LLMs) and designing robust tools and workflows to enable their use in production.
- Deep learning fundamentals, including implementing models from scratch using NumPy for mathematical understanding.

Programming languages & Frameworks:

- Python
- Elasticsearch
- Django
- Pytest
- C
- MongoDB
- Flask
- Kubernetes
- S3
- Docker
- GCP
- regex

Publications

- Alhelli, L., **Al-Yafeai, M.**, Daoud, M. S., Elamsy, T. *An Intelligent Route Finder for UAE Desert Drivers Based on the A* Algorithm*. Proceedings of the **8th International Conference on IoT: Systems, Management and Security (IOTSMS)**, Gandia, Spain, 2021, pp. 1–5. <https://doi.org/10.1109/IOTSMS53705.2021.9704931>
- Alabbasi, N., Al-Badrashiny, M., Aldahmani, M., AlDhanhani, A., Alhashmi, A. S., Alhashmi, F. A., Al Hashemi, K., Alkhobbi, R. E., Al Maazmi, S. T., **Alyafeai, M. A.**, Alzaabi, M. M., Alzaabi, M. S., et al. *Gulf Arabic Diacritization: Guidelines, Initial Dataset, and Results*. Proceedings of the **Seventh Arabic Natural Language Processing Workshop (WANLP)**, Abu Dhabi, 2022. <https://doi.org/10.18653/v1/2022.wanlp-1.33>
- Daoud, M. S., Abu Shawar, B., **Alyafeai, M.**, Essam, A., Ghassan, M., Alzaabi, N. *Enhancing Intrusion Detection Systems Accuracy Using Machine Learning*. Proceedings of the **International Conference on Software Defined Systems (SDS)**, 2023, pp. 103–106. <https://doi.org/10.1109/SDS59856.2023.10329276>
- Malartic, Q., Roy Chowdhury, N., Cojocar, R., Farooq, M., Campesan, G., Djilali, Y. A. D., Narayan, S., Singh, A., Velikanov, M., Boussaha, B. E. A., **Al-Yafeai, M.**, Alobeidli, H., Al Qadi, L., et al. *Falcon2-11B Technical Report*. arXiv preprint, 2024. arXiv:2407.14885
- Boussaha, B. E. A., **Alyafeai, M.**, Alzubaidi, A., Al Qadi, L., Alsuwaidi, S., Hacid, H. *Adapting Falcon3-7B Language Model for Arabic: Methods, Challenges, and Outcomes*. Proceedings of the **Third Arabic Natural Language Processing Conference (ArabicNLP)**, Suzhou, China, 2025, pp. 1–15.
- Boussaha, B. E. A., Al Qadi, L., Farooq, M., Alsuwaidi, S., Campesan, G., Alzubaidi, A., **Alyafeai, M.**, Hacid, H. *3LM: Bridging Arabic, STEM, and Code through Benchmarking*. Proceedings of the **Third Arabic Natural Language Processing Conference (ArabicNLP)**, Suzhou, China, 2025, pp. 42–63.
- *Evaluating Arabic Large Language Models: A Survey of Benchmarks, Methods, and Gaps*.

Certificate & Achievements

- **Release Falcon H1 Arabic (Jan 2026)**
 - Primary contributor after the lead, responsible for the end-to-end continuous pre-training of Falcon-H1 Arabic.
- **Release Falcon Arabic (May 2025)**
 - Primary contributor after the lead for Falcon Arabic, driving data preparation and tokenization through evaluation-driven analysis of model weaknesses and targeted data refinement.
- **WeCloud AI Course (October 2024)**
 - A course for Learning how to preprocess data, train and finetune LLMs
- **Project Management Simplified (Aug 2024)**
 - Project management certificate that introduces the 12 steps for a successful completion of a project.
- **Project Management Foundation (Jul 2024)**
 - Project management certificate that goes through all parts related to project management from an abstract view.
- **Won Sibaq Lahja Program (Oct 2022)**
 - Participated In Sibaq Lahja, and manage to win the first for building a model that diacritize gulf Arabic text.
- **Agile Frameworks Fundamentals (Aug 2022)**
 - Online Course About Different Agile Framework “Udacity”.